Digital Forensics 1
Rubric 3

Digital Evidence Analysis
The project must include extraction of relevant artifacts through file system analysis using Autopsy or FTK. Create a timeline of events based on file timestamps, registry entries, and logs. Recover deleted files using data carving with Scalpel or Foremost. Recover deleted data using specialized forensic tools. Document findings with screenshots, methodology explanations, and artifact correlation establishing factual timelines.

To satisfy the **Digital Evidence Analysis** section of your rubric, the report must show a transition from evidence collection to a rigorous, deeply analytical investigation. This phase proves *what* happened, *when* it happened, and *who* did it.

The core of this phase is artifact correlation: matching a carved file to a registry key, linking it to an OS event log, and placing every single finding into a unified, chronological timeline.

Here is the step-by-step breakdown of how to execute these analytical requirements and document them to max out your rubric score.

🔍 1. File System Analysis & Artifact Extraction

Using **Autopsy** or **FTK**, you need to parse the file system to extract evidence that points directly to the incident (e.g., data exfiltration, policy violation, or malware execution).

- **What to Look For & Extract:**
    - **User Activity:** Examine the target user's directory (`\Users\[Username]\Downloads`, `\Desktop`, `\Documents`).
    - **Web Artifacts:** Extract browser histories, cookies, and cache files (Autopsy automatically parses these under the "Web History" tree module).
    - **System Files:** Look for hidden folders, unauthorized software installations, or staging folders where data was gathered.

- **Documentation Requirement:** Document your **methodology**. Explain *why* you searched specific paths. Take screenshots of the extracted artifacts highlighted inside the tool's interface, showing their file paths and metadata.

⛏️ 2. Data Carving (Recovering Deleted Files)

When a file is deleted, its file system pointers are erased, but the raw data remains in unallocated space. Your rubric explicitly requires recovering these using **Scalpel** or **Foremost**, alongside specialized forensic suites.

- **Execution (Scalpel/Foremost in Linux/SIFT Workstation):**
    - Point the tool at your raw image file (`.dd` or `.raw`) or a specific unallocated space extract:
        - *Command:* `foremost -t jpg,pdf,doc -i evidence_image.dd -o /output/recovered_files/`
        - *Command:* `scalpel -c scalpel.conf -o /output/recovered/ evidence_image.dd`
    - Check the output directory for successfully reconstructed headers and footers.
- **Execution (Autopsy/FTK):**
    - Autopsy natively runs data carving via the **PhotoRec Ingest Module**. Navigate to the **"Deleted Files"** virtual folder in the left tree menu to view carved files.
- **Documentation Requirement:** Take a screenshot of the carving command running or the "Deleted Files" panel in Autopsy. You must include a brief **methodology explanation** explaining how file headers/footers (magic bytes) allow carving tools to reconstruct data without file system metadata.

⏰ 3. Creating a Unified Timeline of Events

A disorganized list of files will fail the rubric. You must synthesize your findings into a single, cohesive **Master Timeline** sorted chronologically.

- **Timeline Sources to Correlate:**
    - **File Timestamps:** MACB times (Modified, Accessed, Created, Born).
    - **Registry Entries:** `LastWrite` times of critical keys (e.g., *ShellBags* showing folders opened, *UserAssist* showing programs executed, *USBSTOR* showing when an external drive was plugged in).
    - **Windows Event Logs:** Security logs (Event ID `4624` for successful logins), System logs, or Application logs matching the times of file modifications.
- **Documentation Layout:** Present this as a highly scannable, clean data table.

Timestamp (UTC / Local)	Artifact Source	Event Type	Description & Forensic Significance	Artifact Correlation / Proof
2026-10-29 08:12:14	`SYSTEM` Registry	USB Connect	USB device (Vendor: SanDisk, S/N: 12345) connected to the system.	*Correlates with malicious data transfer window.*
2026-10-29 08:15:30	File System (MFT)	File Modify	Sensitive file `Quarterly_Report.xlsx` accessed and copied.	*MACB properties show "Last Accessed" matches USB connection.*
2026-10-29 08:22:00	Unallocated Space	File Deleted	File `Quarterly_Report.xlsx` deleted by user.	*Carved via Foremost; recovered document confirms content match.*
2026-10-29 08:23:45	`Security.evtx`	User Logout	User account `jsmith` logs out of the workstation.	*Event ID 4647 confirms session termination*

	immediately post-deletion.

📝 Step-by-Step Reporting Methodology Checklist

To guarantee compliance with the rubric's documentation constraints, ensure every major analytical finding in your report includes these three components:

1. **Methodology Explanation:** A paragraph explaining *how* the tool parsed the data (e.g., *"Autopsy was utilized to parse the $MFT to extract MACB timestamps..."*).
2. **Visual Anchor (Screenshot):** A clear screenshot showing the artifact, its file path, and its metadata properties window.
3. **Artifact Correlation:** A text description directly linking the artifact to another piece of evidence (e.g., *"The creation time of the carved JPEG file recovered from unallocated space matches exactly with the volume shadow copy timestamp from..."*).